



Arin Kumar Singh

Roll No.:220107016

B.Tech - Chemical Engineering

Indian Institute Of Technology, Guwahati

+91-6202071104

s.arin@iitg.ac.in

arinsingh012@gmail.com

LinkedIn | GitHub

EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. Major	Indian Institute of Technology, Guwahati	8.0(Current)	2022-Present
Senior Secondary	CBSE Board	97.2%	2021
Secondary	CBSE Board	95.2%	2019

EXPERIENCE

- Dyet Nutrition** May 2025 - July 2025
Data Science Intern Remote
 - Developed a personalized **AI chatbot** leveraging a **multi-modal RAG pipeline** for research articles and clinical studies.
 - Built document parsing workflows with **Unstructured** and **Chroma DB**, using **GPT-4o** to process **10,000+** PDFs into **1M+ embeddings** to generate precise text summaries, improving retrieval accuracy and answer comprehensiveness.
 - Developed robust **pytest** based automated test suites to validate AI-driven document extraction workflows for **5000+** PDFs.

PROJECTS

- Forest Fire Prediction & Simulation** June 2025 - July 2025
Bharatiya Antariksh Hackathon 2025 - ISRO GitHub
 - Built a **shallow U-Net** with **13 geospatial layers** and **ComboLoss (BCE+IoU)** for next-day wildfire probability mapping.
 - Addressed **class imbalance** and improved predictions with **post-processing techniques** yielding geographically valid maps.
 - Predicted per-pixel wildfire **Rate of Spread** using a hybrid **CNN+S-LSTM** architecture with **physics-based corrections**.
 - Modeled wildfire dynamics with **Probabilistic Cellular Automata**, simulating spread across 1–12 hour horizons.
- PDF Question Answering System** June 2024
Freelance Project GitHub
 - Designed and developed an interactive **Streamlit web application** for answering questions about PDF content.
 - Built a **RAG pipeline** using Sentence BERT **embeddings** and **cosine similarity**, achieving **83%** document **retrieval accuracy**.
 - Integrated **Hugging Face Transformer models** to generate context-aware answers from retrieved documents.
 - Managed **data preprocessing** and **optimization for rapid retrieval** and real-time question answering capabilities.
- Amex Offerings Personalization** July 2025
American Express Campus Challenge 2025 GitHub
 - Built an **offer ranking system** on **770K+** customer interactions using **LightGBM** with **LambdaRank** for personalized relevance.
 - Boosted **MAP@7** from **0.21** to **0.53** using **5:1 negative sampling** and **51+** temporal, behavioral, semantic & spending features.
 - Tuned with **Optuna (850+ trials)** and **GroupKFold validation**, using **SHAP** for interpretability and consistent performance.

TECHNICAL SKILLS

- Programming:** C/C++, Python, MATLAB*
- Data Analysis:** Pandas, NumPy, Matplotlib, Seaborn
- Database Management:** MySQL, PostgreSQL
- Web Technologies:** HTML, CSS
- ML Frameworks:** Scikit-learn, PyTorch, TensorFlow*

*Elementary Proficiency

KEY COURSES TAKEN

- Mathematics:** Linear Algebra, Basic Calculus, Differential Equations, Complex Analysis, Complex PDE
- Computer Science:** Introduction to Computing, Computing Laboratory, Computer Aided Numerical Methods
- Data Science:** Application of AI and ML in chemical engineering, Summer Analytics (Data Science Bootcamp)

POSITIONS OF RESPONSIBILITY

- Overall Coordinator**, Student Welfare Board, IIT Guwahati April 2024 - April 2025
 - Led a **three-tier** team of **500+** members across **7** clubs, inter-hostel competition and a fest, impacting **8,000+** students.
 - Allocated a consumable budget of **INR 40Lac**, executed **100+** **events**, partnered with **30+** NGOs, driving **5x** scalability.
 - Spearheaded Prayatna, **North East's largest student led socio welfare initiative**, impacting **20K+** attendees nationwide.

ACHIEVEMENTS AND EXTRA-CURRICULARS

- Amazon ML Summer School 2024**, Selected for the flagship Machine Learning program among 80,000+ applicants. 2024
- ISRO BAH 2025**, Ranked among the Top 10 in our Problem Statement out of 8,744 participating teams. 2025
- LeetCode**, Achieved a peak rating of 1832, placing in the top 6.4% of global participants. 2025
- Bronze Medal, Startup Competition**, Led a team of 15 to ideate, execute, and pitch a startup among 14 hostels. 2024
- InnovateX, E-Cell IITB**, Secured 9th place in national case study competition among 2,200+ participants. 2024
- Organizer, Techniche**, Organized Asia's first-of-its-kind AquaWar event with a budget of over 50+ lakh. 2023
- JEE Mains 2022**, Secured 99.43 percentile among 870,000+ candidates nationwide. 2022
- JEE Advanced 2022**, Ranked 5339 among 160,000+ candidates nationally. 2022